Physics-Inspired Machine Learning for Quantum Error Mitigation

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Noise is a major obstacle in current quantum computing, and Machine Learning for Quantum Error Mitigation (ML-QEM) promises to address this challenge, enhancing computational accuracy while reducing the sampling overheads of standard QEM methods. Yet, existing models lack physical interpretability and rely heavily on extensive datasets, hindering their scalability in large-scale quantum circuits. To tackle these issues, we introduce the Neural Noise Accumulation Surrogate (NNAS), a physics-inspired neural network for ML-QEM that incorporates the structural characteristics of quantum noise accumulation within multi-layer circuits, endowing the model with physical interpretability. Experimental results demonstrate that NNAS outperforms current methods across a spectrum of metrics, including error mitigation capability, quantum resource consumption, and training dataset size. Notably, for deeper circuits where QEM methods typically struggle, NNAS achieves a remarkable reduction of over half in errors. NNAS also demands substantially fewer training data, reducing dataset reliance by at least an order of magnitude, due to its ability to rapidly capture noise accumulation patterns across circuit layers. This work pioneers the integration of quantum process-derived structural characteristics into neural network architectures, broadly enhancing QEM's performance and applicability, and establishes an integrative paradigm that extends to various quantum-inspired neural network architectures.

I. INTRODUCTION

Quantum computing stands as a groundbreaking advancement that can tackle problems beyond the reach of classical computing, and thus revolutionizes fields like cryptography, material science, and optimization. Despite these advances, quantum noise, stemming from environmental interactions and hardware imperfections, poses a threat to the reliability and scalability of quantum computing [1–4]. In response, Quantum Error Mitigation (QEM) [5–9] has become a critical field, offering qubit-conserving strategies to mitigate the noise impact on measurement results without the need for millions of qubits required by full quantum error correction [10–16], thus expediting near-term quantum computing applications. Standard QEM techniques, such as Zero-Noise Extrapolation (ZNE) [17–20], Probabilistic Error Cancellation (PEC) [21–23], and Clifford Data Regression (CDR) [24–26], can effectively reduce errors but face scalability issues due to their high sampling overheads [27, 28], which scale up to exponentially with the circuit's gate number [27]. This exponential growth in sampling requirements poses significant hurdles for the practical application of QEM, particularly for complex quantum algorithms that involve deeper circuits.

Within this landscape, Machine Learning for Quantum Error Mitigation (ML-QEM) [29–33] stands out for its ability to retrieve noiseless results from noisy measurement data utilizing statistical model learning. In this manner, ML-QEM substantially cuts down on sampling overhead, thus economizing on quantum resources compared to standard QEM. Recently, through comparative benchmarking of prevailing data-driven ML models [29]—encompassing random forests (RF), multi-layer perceptrons (MLP), and graph neural networks (GNN)—across a spectrum of quantum tasks, ML-QEM has demonstrated significant outperformance over the standard QEM approaches with reduced sampling overheads. However, these conventional data-driven models, which are designed without specific physical context, empirically extract the intricate noisy-to-noiseless mappings. This learning pattern not only limits the physical interpretability of the models but also makes their efficacy highly dependent on the quantity and quality of the datasets. As quantum systems scale up, collecting data that meets such high standards becomes increasingly challenge, whether using real quantum devices or classical simulators. Notably, classical simulators are often relied upon to obtain the noiseless outputs of quantum circuits, which are then used as labels. Yet, the complexity of classical simulations required to generate noiseless outputs within quantum datasets grows exponentially with the system size, and this complexity can be further amplified, especially as the entanglement between quantum gates increases [17, 34].

To address these challenges, we've developed a paradigm that deeply integrates QEM with artificial intelligence, taking a pioneering step towards tailoring machine learning models

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for QEM with physical priors. We introduce the Neural Noise Accumulation Surrogate (NNAS), a physics-inspired neural network designed for mitigating errors in complex quantum circuits. NNAS's modules are engineered to capture the cumulative effects of noise across quantum circuit layers, offering theoretical interpretability, and its alignment with physical processes is validated by experimental visualizations. This capability enables NNAS to emulate noise accumulation and predict its impact on measurements effectively. Our experiments demonstrate that NNAS maintains the strengths of ML-QEM, achieving state-of-the-art (SOTA) error mitigation with minimal quantum sampling overhead. It also overcomes the data-intensive nature of previous ML-QEM methods, training effectively with a limited dataset. Moreover, NNAS's proficiency in learning noise patterns enables effective error mitigation even in deeper circuits, a task typically difficult for QEM due to signal-to-noise challenges. Consequently, NNAS significantly enhances ML-QEM's capacity to handle complex quantum circuits, bolstering the utility of near-term quantum devices.

II. NEURAL NOISE ACCUMULATION SURROGATE FOR ERROR MITIGATION

To facilitate a general analysis, we model the quantum algorithms based on multi-layer circuits as layer-wise quantum sequential tasks, as depicted in Fig.1 **a**. We examine the quantum circuit with L layer of gates, denoted as $\mathcal{U}_L = u_L \circ u_{L-1} \circ ...u_1 = \prod_{l=1}^{L} u_l$, under the canonical Completely Positive and Trace-Preserving (CPTP) [35] noise postulate. The noisy quantum circuit, denoted as $\tilde{\mathcal{U}}_L$, can be expressed as the product of L noisy layers: $\tilde{\mathcal{U}}_L = \prod_{l=1}^{L} \tilde{u}_l$. Each noisy layer is given by $\tilde{u}_l = \mathcal{E}_l \circ u_l$, where $\mathcal{E}_l$ represents the layer-dependent CPTP noise channel.

In an effort to embed physical insights into the noisy-to-noiseless mapping that complements data-driven learning, we craft a model which harnesses layer-wise sequence data during training stage to capture the characteristics of noise accumulation patterns. Beyond mere sequence processing, our model is adaptable, capable of yielding output from a specific layer during prediction. This can be achieved by treating the output as a sequence and focusing on the target layer's result. Hence, we formulate the objective of the mitigation process by recovering the noiseless result sequences $\{y_l\}_{l=1}^{L}$ from noisy sequence $\{\tilde{y}_l\}_{l=1}^{L}$.

We then dissect the propagation of noise, aiming to isolate the noise impact to distill its cumulative effect within the circuit. By relocating each layer's noise channel to the end of circuit, we derive the noisy output state ρ_l for the l-th layer:

$$\tilde{\rho}_l = (\prod_{j=1}^{l}(1-p_j))\rho_l + \left(\mathcal{U}_l \mathcal{N}_l \mathcal{U}_l^{\dagger}\right)\rho_l, \tag{1}$$

with the effective rate p_j and cumulative noise $\mathcal{N}_l$ derived from maximum noise decomposition (MND) of $\mathcal{E}_l$ (More details in 'Methods'). We define $R_l = \mathcal{U}_l \mathcal{N}_l \mathcal{U}_l^{\dagger}$ as the noise envelope and quantify its impact on results through the impact factor r_l. Given an observable O, the task of recovering noiseless results is translated into the estimation of effective rates $\{\hat{p}_j\}_j$, which are the initial values obtained prior to mitigation (further details in Supplementary Section 1), and the impact factor $\hat{r}_l$ which is the output of our model. The mitigated expectation value is then calculated using the formula:

$$y_l^{\text{em}} = \frac{\tilde{y}_l}{\prod_{j=1}^{l}(1-\hat{p}_j) + \hat{r}_l}. \tag{2}$$

Isolating the noise envelope R_l offers two significant advantages. First, it sharpens our focus on characterizing noise by directly assessing its impact on final results. Second, the estimated value $\prod_{j=1}^{l}(1-\hat{p}_j)$ can guide the trend of mitigated values across layers, aligning initial network training more closely with the target. This enhances the model's efficiency in terms of training resources and robust performance, particularly in deeper layers, as empirically validated in the subsequent 'Results' section.

We refine our understanding of R_l by developing a physical process framework that delineates its layer-by-layer dynamics, which serves as the foundational structure for our NNAS. This framework is segmented into three tiers, capturing the critical components of noise accumulation: input, noise accumulation function, and noise impact extraction. The NNAS is designed in accordance with this framework, aligning its modules with the framework's tiers, and it predicts the estimation of r_l for the mitigated result calculation as given in Eq. 2. The overall pipeline is depicted in Fig.1 **b**.

A. Uniform neural accumulator based on layer-wise hidden state

The noise envelope R_l consists of two components: the quantum circuit $\mathcal{U}_l$ and cumulative noise $\mathcal{N}_l$, forming the functional basis for noise accumulation. We reformulate them recursively in terms of the layer depth by

$$\mathcal{U}_l = u_l \circ \mathcal{U}_{l-1} = \phi_{\mathcal{U}}(\mathcal{U}_{l-1}, u_l), \tag{3}$$

and

$$\begin{aligned}\mathcal{N}_l &= (1-p_l)\mathcal{N}_{l-1} + p_l \prod_{j=1}^{l-1}(1-p_j)a_l + p_l a_l \circ \mathcal{N}_{l-1} \\ &= \phi_{\mathcal{N}}(\mathcal{N}_{l-1}, \mathcal{U}_{l-1}, \tilde{u}_l),\end{aligned} \tag{4}$$

respectively, to capture their progressive intricacies. Here $a_l = \mathcal{U}_l^{\dagger}\Lambda_l\mathcal{U}_l$ denotes the noise introduced by the l-th noisy layer and Λ_l is the effective noise derived from MND of $\mathcal{E}_l$. We collectively term these recursive relations as the noise accumulation function ϕ, synthesized from $\phi_{\mathcal{U}}$ and $\phi_{\mathcal{N}}$. The amalgamated input to ϕ is the noisy layer $\tilde{u}_l = \mathcal{E}_l \circ u_l$.

To map these dynamics, we employ feature embedding on M input variables, capturing the physical traits of the noisy layer and standardizing multi-structured quantum data into embedded data $X \in \mathbb{R}^{M*L}$, which serves as the input for RNN. To enhance model robustness, especially with limited

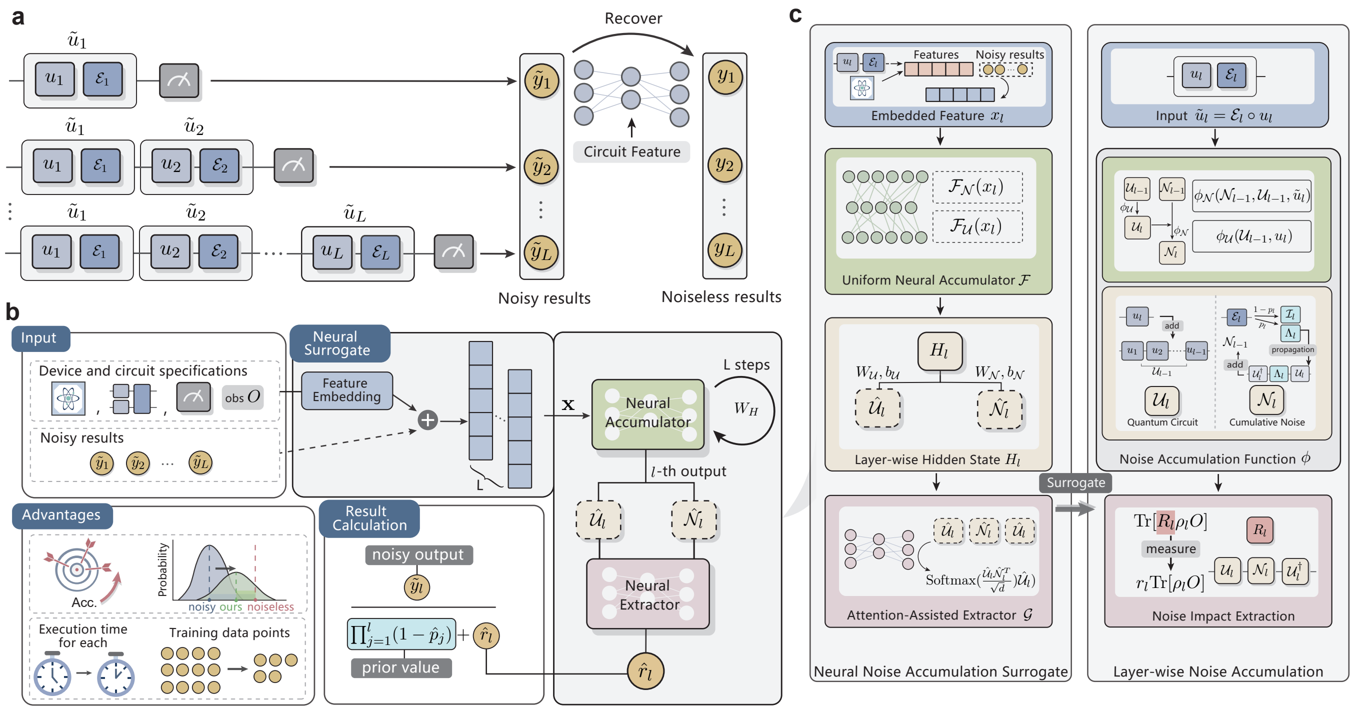

FIG. 1. **The architecture of the physical-inspired machine learning for quantum error mitigation.** **a** The illustration of layer-wise quantum sequential task. We model the quantum algorithms based on multi-layer circuits as quantum sequential tasks for a general analysis. The noisy quantum circuit comprises L layers of gates $\tilde{\mathcal{U}}_L = \tilde{u}_L \circ ... \circ \tilde{u}_2 \circ \tilde{u}_1$, with each noisy layer $\tilde{u}_l$ decomposed into the noiseless layer u_l and the noise channel $\mathcal{E}_l$, yielding the noisy results $\{\tilde{y}_l\}_l$ by measurement at each layer depth. We can recover the noiseless values $\{y_l\}_l$ from $\{\tilde{y}_l\}_l$ via machine learning (ML) for quantum error mitigation (QEM), with a neural network fed by descriptive features of circuits. **b** The overall pipeline of our Neural Noise Accumulation Surrogate (NNAS) for error mitigation. The input data including descriptive features for device and circuit specifications, and optional noisy result sequences are fed into NNAS, which encompasses three successive stages: embedding, neural accumulator, and neural extractor. The output of NNAS is the prediction of noise impact factors $\{\hat{r}_l\}_l$ for subsequent result calculation. The NNAS offers two key advantages: enhanced accuracy of mitigated values and reduced quantum resource requirements for training dataset. **c** Mapping relationship for constructing NNAS which conforms to the Layer-wise Noise Accumulation Framework designed for physical processes. The physical framework encompasses three components, which is mapped into NNAS as Embedded Feature, Uniform Neural Updater $\mathcal{F}$ along with the layer-wise hidden state H_l, and the Attention-Assisted Extractor $\mathcal{G}$. The embedded feature x_l is obtained from the embedding of specification features, optionally incorporating noisy results. Within the Noise Accumulation Function tier of the physical framework, the cumulative noise N_l at the l-th layer is updated based on $\mathcal{N}_{l-1}$. This update incorporates the twirled effective noise Λ_l relocated to the end of the circuit $\mathcal{U}_l$, where Λ_l is derived from the layer-dependent noise channel $\mathcal{E}_l$ at l-th layer. In the Noise Impact Extraction, the noise envelope $R_l = \mathcal{U}_l \mathcal{N}_l \mathcal{U}_l^\dagger$ is quantified as the noise impact factor r_l on the noiseless expectation value $\mathrm{Tr}[\rho_l O]$ obtained from the noiseless output quantum state ρ_l and observable O.

descriptive data, we optionally concatenate the noisy results into X, with careful consideration and fine-tuning for specific tasks.

Separate handling of the accumulation functions $\phi_{\mathcal{U}}$ and $\phi_{\mathcal{N}}$ results in loss of interwoven information. Despite the possibility of merging them at each step, this method leads to both error accumulation and prolonged total computation time. To address this, we propose a uniform neural accumulator $\mathcal{F}$, formulated as a recursive neural network (RNN) [36, 37], which integrates both $\mathcal{U}_l$ and $\mathcal{N}_l$ into a layer-wise hidden state H_l. This RNN is designed to capture the layer-depth-relevant dependence inherent in the evolution of both $\mathcal{U}_l$ and $\mathcal{N}_l$, ensuring a conformal mapping to the noise accumulation process that is both effective and computationally efficient. The hidden state H_l is recursively calculated as follows:

$$H_l = \mathcal{F}(X_l, H_{l-1}), \tag{5}$$

where information accumulates progressively with layer depth. This recursive integration within H_l continues until the neural extractor, where we distill the estimated noise impact factor $\hat{r}_l$. We employ learnable layer to recover the key components of $\mathcal{U}_l$ and $\mathcal{N}_l$ as $\hat{\mathcal{U}}_l$ and $\hat{\mathcal{N}}_l$ by

$$\begin{aligned} \hat{\mathcal{U}}_l &= W_{\mathcal{U}} \cdot H_l + b_{\mathcal{U}}, \\ \hat{\mathcal{N}}_l &= W_{\mathcal{N}} \cdot H_l + b_{\mathcal{N}}, \end{aligned} \tag{6}$$

where $W_{\mathcal{U}}$, $W_{\mathcal{N}}$, $b_{\mathcal{U}}$ and $b_{\mathcal{N}}$ are learnable parameters.

B. Attention-assisted noise impact extractor

Aiming to capture the symmetric structure of the noise impact within $R_l = \mathcal{U}_l \mathcal{N}_l \mathcal{U}_l^\dagger$, we employ an attention mechanism tai-

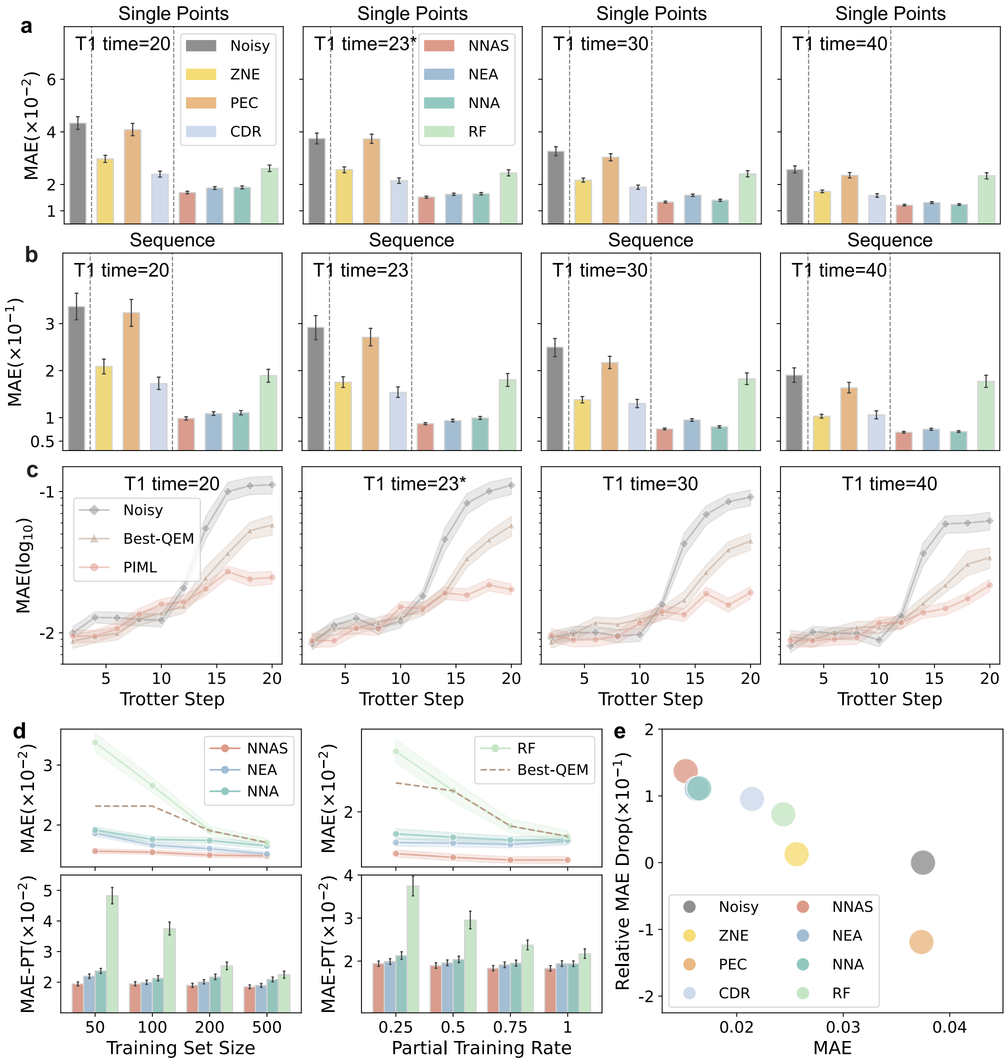

FIG. 2. **Evaluation results on Quantum Approximate Optimization Algorithm-type multi-layer circuits. a-b** The Mean Absolute Error (MAE) of baselines in different scenarios with increasing T1 time. The MAE is calculated using (upper) single points and (lower) sequence-derived vector norms. The Longer T1 time denotes lower noise level. We focus on T1 = 23.235 μs marked with 23*, which represent the state-of-the-art quantum coherence times [3]. **c** The comparison among Noisy, best-performing QEM methods including standard QEM methods and RF (denoted as Best-QEM), our model with growing Trotter step. We assess performance using single-point MAE computations. For **a-c**, we train the models using 100 sequences with partial training rate 0.25. **d** (upper) The MAE of full test set for all the ML-QEM methods with varying training set size and partial training rate. (lower) The MAE-PT, which represents of MAE for partial test set with layer depth 11-20, for ML-QEM methods. **e** The comparison of performance for mitigation during test stage among baselines. The performance metrics of our analysis are characterized by the Mean Absolute Error (MAE) and the drop in relative MAE, with the latter defined as the logarithmic ratio between the MAE of noisy and mitigated expectation values. Notably, a larger relative MAE drop signifies better performance, indicating a more effective mitigation of noise. All the data in the figure presents average results with 95% confidence interval error bars.

lored to align with this symmetry. Given that $\hat{\mathcal{U}}_l$ and $\hat{\mathcal{N}}_l$ are one-dimensional low-rank compressed vectors derived from H_l, directly emulating the matrix form $\mathcal{U}_l\mathcal{N}_l\mathcal{U}_l^\dagger$ is not feasible due to the dimensional constraints. Instead, we leverages an attention mechanism to merge information from the surrogates $\hat{\mathcal{U}}_l$ and $\hat{\mathcal{N}}_l$. The attention mechanism is formulated as follows:

$$\mathrm{A}_l = \mathrm{Softmax}\left(\frac{\hat{\mathcal{U}}_l\hat{\mathcal{N}}_l^T}{\sqrt{d}}\right)\hat{\mathcal{U}}_l = \mathcal{G}(\hat{\mathcal{U}}_l, \hat{\mathcal{N}}_l), \qquad (7)$$

which introduces a computationally symmetric pattern that mirrors the conjugate symmetry of R_l.

The benefits of using the attention mechanism are twofold. First, the Softmax activation enhances the information density within $\hat{\mathcal{U}}_l\hat{\mathcal{N}}_l^T$ by normalizing attention scores, thereby highlighting the most pertinent information. Second, dividing by $\sqrt{d}$ enhances stability in the computation. This approach partially restores the compressed information and effectively fuses the representations from the low-rank compression of $\hat{\mathcal{U}}_l$ and $\hat{\mathcal{N}}_l$, achieving enhanced stability and efficacy by preserving the symmetrical structure with the attention-assisted extractor.

III. RESULTS

We initially evaluate our model's performance on a typical class of multi-layer circuits, with exploration of the alignment of the structural features of the cumulative noise $\mathcal{N}$ and its surrogate $\hat{\mathcal{N}}_l$. Additionally, we validate NNAS's versatility by customizing it for a specific qubit-wise quantum algorithm task. Baselines include the noisy case (Noisy), standard QEM methods (ZNE, PEC, CDR), typical ML-QEM methods using random forest (RF), and our ablated models: the neural extractor ablated (NEA) and the quantum-inspired neural network-both neural accumulator and extractor-ablated (NNA) (See Supplementary Section 2 for baseline details.). All baselines are trained with mean squared error (MSE). We define the *hard regime* in our dataset where sequences exceed half the maximum sequence length, indicating higher complexity in data acquisition. And we denote the proportion of this regime in the training set by the partial training rate p_r. The generation of the dataset is elaborated in Supplementary Section 3, and the detailed experimental settings, evaluation metrics, and extended results are provided in Supplementary Section 4.

A. Evaluations on typical layer-wise quantum circuits—QAOA-type circuits

Our evaluations are centered around Quantum Approximate Optimization Algorithm (QAOA)-type circuits, which are crucial for near-term quantum algorithms, including quantum Trotterized simulations [39–41] and variational quantum algorithms (VQAs) [42–44]. These QAOA-type circuits are distinguished by their layer units, which are determined by a certain Hamiltonian. In our experiments, we focus on the temporal evolution of the 1D transverse-field Ising spin chain [39], as described by the Hamiltonian $H_{\text{Ising}}(J, h)$ with circuit parameters J and h. We utilize the first-order Trotterized circuit, a standard configuration for QAOA-type circuits in quantum Trotterized simulations. All machine learning models are trained on sequence data, and the Mean Absolute Error (MAE) is calculated using both sequence norms and single point methods. This allows us to validate the effectiveness of our model in two applications that leverage QAOA-circuits: quantum Trotterized simulations and VQAs, respectively. We evaluate our model's performance on six-qubit circuits with random circuit parameters, varying the layer depth (Trotter step) from 1 to 20. Our training set encompasses a hard regime including on steps 11-20. In the testing stage, we assess all baselines across 200 sequences, spanning the full range of Trotter steps from 1 to 20.

Result Comparisons. Our model demonstrates superior performance over baseline models across various noise levels, particularly excelling in high-noise scenarios as shown in Fig.2 **a** and **b**. Notably, this achievement comes with training on a limited dataset of 100 sequences at a partial training rate of $p_r = 0.25$, highlighting the efficiency of our model in data usage. Our model significantly reduces MAE by an average of 65.85% across two key calculations compared to the noisy case (Noisy) and by 35.12% over the best-performing QEM methods, including standard QEM and RF approaches. This advantage is accentuated at larger Trotter steps, especially beyond Trotter step 15, where our model achieves a 79.42% MAE reduction compared to Noisy and a 55.63% improvement over the best-performing QEM methods as depicted in Fig.2 **c**. These results underscore the scalability and robustness of our model's theoretical design.

We then explore our model's efficiency compared to baselines using ML models, including RF and our ablated models, under certain noise conditions that reflect SOTA quantum computational capabilities [3]. As given in Fig.2 **d-e**, the former focuses on the training phase, while the latter focuses on the prediction phase. In Fig.2 **d**, while RF exhibits a pronounced sensitivity to changes in dataset scale, NNAS demonstrates robustness to this variations. This is evidenced by its ability to maintain accuracy with a 90% reduction in dataset size and a 75% decrease in p_r, along with an order-of-magnitude reduction in the total number of data points across all sequence data. As depicted in Fig.2 **e**, we demonstrate the overall performance for mitigation among baselines during prediction, where our model achieves peak accuracy, emphasizing its effectiveness and efficiency.

Explorations on how neural networks surrogate the cumulative noise. The comparative results above, including results of ablated models NEA and NNA with an average MAE reduction of 8.58% over NEA and 6.99% over NNA, preliminarily confirm that the synergistic integration of the nueral noise accumulator and extractor into our model significantly enhances performance. Building on these findings, we will further substantiate our model's enhanced performance by delving into its underlying mechanism. We aim to investigate the structural alignment between the quantum cumulative noise $\mathcal{N}$ and its surrogate $\hat{\mathcal{N}}$, which is crucial for assessing the model's ability to embed structural information. Our analysis begins with a case study, as given in Fig.3 **a-c**. Fig. 3 **a** presents the structural evolution of $\mathcal{N}$ and $\hat{\mathcal{N}}$, visualized using the Pauli Transfer matrix [38] (PTM) for $\mathcal{N}$ with compression via sum pooling and the two-dimensional matrix $\hat{\mathcal{N}}^T\hat{\mathcal{N}}$ for $\hat{\mathcal{N}}$. With increasing Trotter steps, the distribution across these matrices shows increased dispersion, indicating a progressive alignment of structural changes between $\mathcal{N}$ and $\hat{\mathcal{N}}$. Fig. 3 **b** extends the analysis by computing the Spearman rank correlation matrix between $\hat{\mathcal{N}}^T\hat{\mathcal{N}}$ and sub-matrices of $\mathcal{N}$. The initial block diagonal-like pattern in the correlation matrix, due to sparse off-diagonal elements of $\mathcal{N}$, transitions to a rapid divergence in correlation values from steps 1 to 3, highlighting the model's swift adaptation to early structural changes. This is followed by a decrease in local correlation, emphasizing the model's focus on global characteristics of $\mathcal{N}$ over specific details, especially as dispersion gradually increases in later steps. Building on this, Fig.3 **c** demonstrates the alignment of differential entropy curves between $\mathcal{N}$ and $\hat{\mathcal{N}}$ across increasing Trotter steps, using differential entropy to gauge the di-

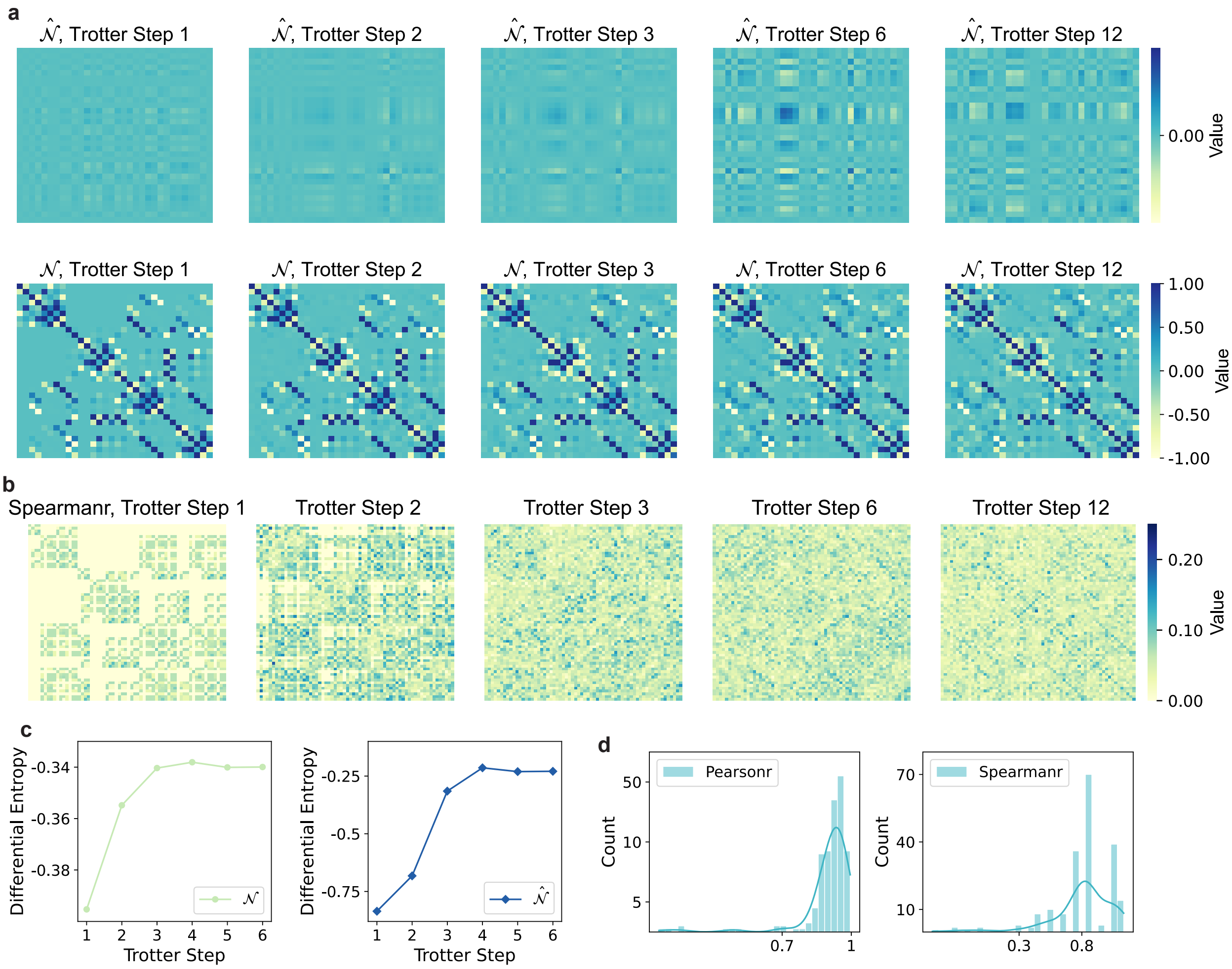

FIG. 3. **Comparative visualization and statistical analysis for the structural correspondence between quantum cumulative noise and its surrogate.** **a** The visualization of the matrix of surrogate $\hat{\mathcal{N}}^T\hat{\mathcal{N}}$ (upper) and the representation of the cumulative noise $\mathcal{N}$ using the compressed Pauli Transform Matrix [38] (PTM) with sum pooling (lower). **b** The visualization of the Spearman correlation matrix illustrates the relationships between the surrogate $\hat{\mathcal{N}}$ and the compressed PTM of $\mathcal{N}$. The values of each matrix are computed using the vectors of $\hat{\mathcal{N}}$ and the corresponding windows in the compressed PTM of $\mathcal{N}$ that match the shape of $\hat{\mathcal{N}}^T\hat{\mathcal{N}}$. **c** The curve of differential entropy for the surrogate $\hat{\mathcal{N}}$ and the compressed PTM of $\mathcal{N}$ with growing Trotter steps. Experiments **a**-**c** constitute a case study on a single sample instance. The observed increase in disorder, mirrored by the corresponding trend in $\hat{\mathcal{N}}$, indicates the escalating effect of cumulative noise on system dynamics, thereby demonstrating the effectiveness of our model in capturing these structural characteristics. **d** The probabilistic distribution of the Pearson (left) and Spearman (right) correlation coefficients between the entropy curves of $\mathcal{N}$ and its surrogate $\hat{\mathcal{N}}$, evaluated across 200 sequences in the test set.

versity within the noise profiles. Expanding on these insights, Fig.3 **d** conducts a statistical evaluation across 200 test sequences, affirming the model's consistent replication of quantum noise's structural dynamics. High average correlation coefficients—Pearson at 0.8991 and Spearman at 0.8028—calculated from the entropy curves of $\mathcal{N}$ and $\hat{\mathcal{N}}$, substantiate the broad validity of our findings. Collectively, these analyses confirm the model's proficiency in capturing the structural nuances of the physical process.

B. Extending to qubit-wise quantum circuits

In this extended experiment, we showcase NNAS's broad applicability by adapting it to specific qubit-wise quantum algorithms, which focus on their performance enhancement as the qubit number scales up. NNAS can be adapted to scenarios where the configuration of initial qubits remains unchanged as additional qubits are integrated. In such cases, we can regard the growth in the number of qubits as equivalent to the growth in layers, treating the maximum qubit count as the upper limit for layer-wise expansion within the existing model framework. To exemplify this extended application, we turn to quantum metrology (QM) [45], a domain where the preci-

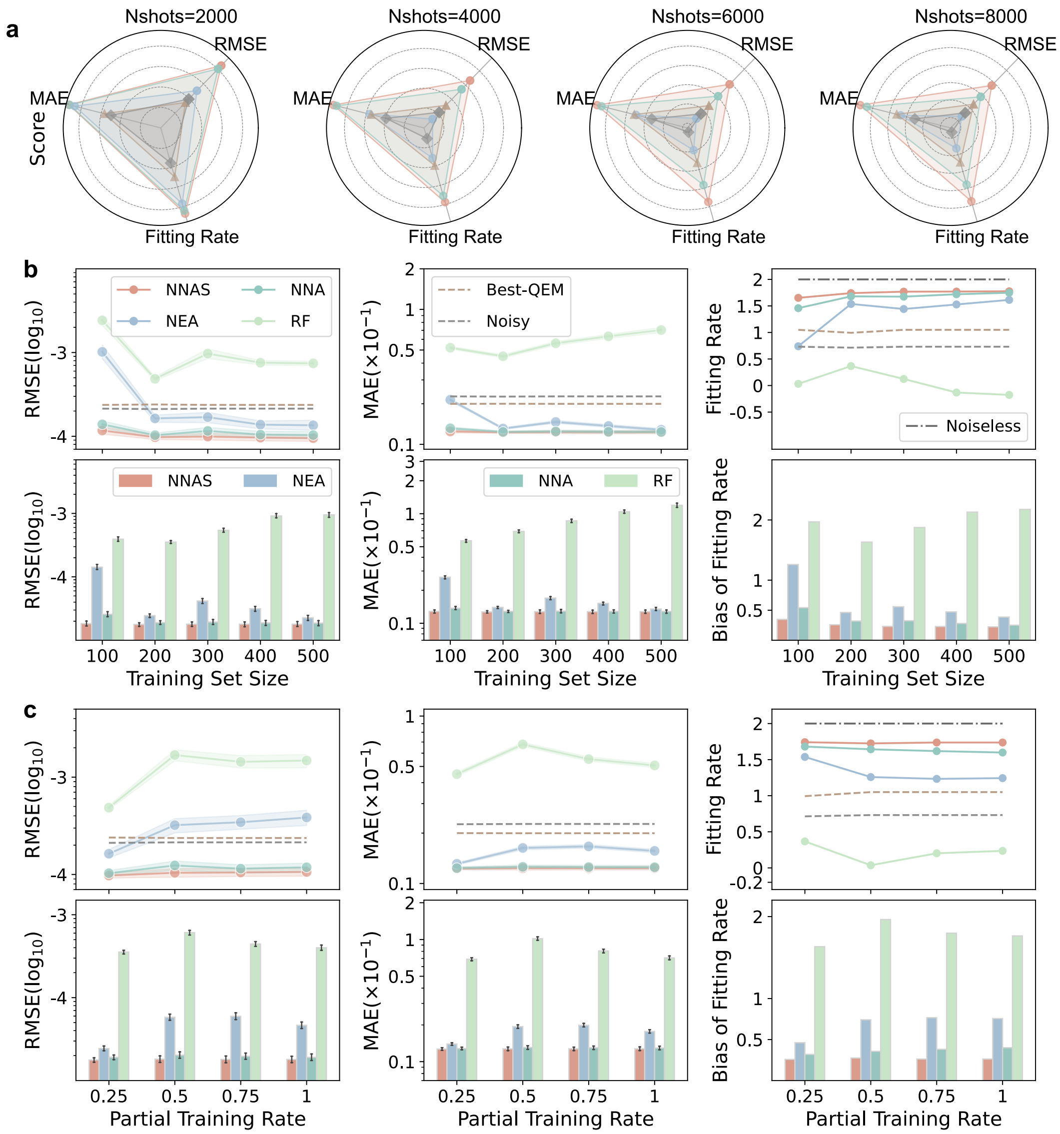

FIG. 4. **Evaluation results on quantum metrology (QM) based on GHZ probe state.** QM focuses on the enhancement of measurement sensitivity as the qubit number scales up. We employ three critical metrics for QM, including the root mean square error (RMSE) for the estimated parameter θ, the MAE for the mitigated expectation value, and the fitting rate of the RMSE scaling curve with respect to qubit number. The RMSE and fitting rate serve as indicators for QM, with the latter being an upper-bounded metric where values closer to 2 signify better performance, as they approach the noiseless scenario. The MAE, alongside the RMSE, quantifies the error in our estimates. **a** The score of three metrics to evaluate the performance on QM. For RMSE and MAE, the score represents the ratio of the noiseless value to the mitigated value, given a finite number of measurement shots. Conversely, for the fitting rate, the score is the ratio of the mitigated value to the noiseless value. We train the ML models on training set with 200 sequences, including GHZ states of 1-10 qubits with hard regime including number of qubits 6-10 and partial training rate 0.25. **b**-**c**, (upper) The metrics of full test set for ML-QEM methods with varying training set size and partial training rate. (lower) The metrics of partial test set with qubit number 6-10 (for RMSE and MAE), and the bias of fitting rate between the mitigated value and noiseless value. All the data in the figure presents average results with 95% confidence interval error bars.

sion of quantum measurements is directly linked to the number of qubits. We employ the Greenberger-Horne-Zeilinger (GHZ) entangled state as the probe state for QM. The circuit for preparing an n-qubit GHZ state is expressed as follows:

$$U_n^{\mathrm{GHZ}} = \prod_{i=1}^{n-1} \mathrm{CNOT}_{i,i+1} = U_{n-1}^{\mathrm{GHZ}} \cdot \mathrm{CNOT}_{n-1,n}, \quad (8)$$

where $\mathrm{CNOT}_{i,i+1}$ denotes the CNOT gate with the i-th qubit as control and the $(i+1)$-th qubit as target. We leverage the structural equivalence of the GHZ state preparation circuit, where each CNOT gate corresponds to a layer in the circuit, directly relating the number of layers to the number of qubits. For each circuit parameter θ, we evolve the GHZ states $\{\mathrm{GHZ}_n\}_n$ under the unitary operations $\{U(\theta)_n\}_n$ and measure them with the global Pauli X observables $\{X^{\otimes n}\}_n$.

Our dataset includes GHZ states with 1 to 10 qubits, focusing on the hard regime of qubits 6-10. Model training involves sequences parameterized by circuit parameters randomly sampled between 0.1 and 5.0 with 0.1 increments, with 10 independent trials per parameter. Evaluations use parameters from 0.05 to 5.05 in 0.1 steps.

To evaluate our model's performance, we employ three critical metrics for QM, including the root mean square error (RMSE) for the estimated parameter θ, the MAE for the mitigated expectation value, and the fitting rate of the RMSE scaling curve with respect to qubit number. The RMSE and fitting rate serve as indicators for QM, with the latter being an upper-bounded metric where values closer to 2 signify better performance, as they approach the noiseless scenario. The MAE, alongside the RMSE, quantifies the error in our estimates. A comparative analysis of these metrics across various measurement shot numbers N_s for baselines is presented in Fig. 3 **a**. Despite trained on a limited dataset of 200 sequences derived from just 20 distinct angles with $p_r = 0.25$, our model significantly outperforms Noisy by 52.81% to 84.95% and the best-performing QEM methods by 46.92% to 53.43% across three key evaluation metrics. Figs. 3 **b** and **c** demonstrate that our model achieves precision advantages with a limited dataset scale compared to other baselines using ML models. Notably, our model realizes a nearly two orders of magnitude reduction in RMSE and a one order of magnitude reduction in MAE relative to the RF model, underscoring the superior efficiency of NNAS in the extended application for mitigating qubit-wise quantum algorithms.

IV. DISCUSSION

Our proposed NNAS demonstrates SOTA performance in results' error mitigation, quantum sampling overhead, and scalability of training datasets, significantly outperforming previous QEM methods. This advancement is attributed to our approach, which goes beyond the conventional ML-QEM by not just employing established machine learning models for error mitigation, but by harmoniously integrating the structural characteristics of quantum noise accumulation within the neural network's architecture. This integration endows our network with physical interpretability and reduces the reliance on the size and scope of quantum data sets. When scaling to large quantum systems, the acquisition of training labels typically requires resource-intensive classical simulations. Although this issue extends beyond our current scope, our framework inherently supports potential solutions. A hybrid strategy (see Supplementary Section 5) combining either (i) near-Clifford circuits [24, 31] or (ii) shallow quantum circuits with low-weight Pauli observables [46] has been shown to maintain computational tractability while preserving error mitigation efficacy.

It's important to note that our focus on multi-layer circuits is universally applicable in quantum computing, meaning our NNAS is nearly adaptable to all current quantum circuits. This significantly enhances the capabilities of QEM, enabling near-term quantum devices to handle larger-scale and deeper circuits, thus paving the way for practical applications.

Moreover, our contribution transcends the realm of QEM; it represents a paradigm that integrates neural network architecture with the evolution of physical states. Our framework for physical processes encapsulates quantum noise accumulation across three dimensions: input, state transition, and output. This multi-tiered structure not only accommodates quantum noise accumulation but also aligns with the overarching principles of physical state evolution, underscoring its adaptability as a general paradigm for physical processes that extend beyond the purview of QEM. Our study highlights the potential of tailoring ML network structures for specific quantum tasks, thereby paving the way for future research in quantum technologies that can achieve high performance with limited quantum datasets.

V. METHODS

A. Layer-wise quantum noise accumulation

The physical framework for layer-wise quantum noise accumulation is established on the basis of the maximum decomposition of layer-dependent noise and the formulation of noise accumulation. We will now detail their derivation as follows.
Maximum noise decomposition. For an n-qubit layer-dependent CPTP noise channel $\mathcal{E}_l$ for noisy layer $\tilde{u}_l$, we define the maximum noise decomposition (MND),

$$\mathcal{E}_l = (1 - p_l)\mathcal{I} + p_l \Lambda_l, \tag{9}$$

with Λ_l representing the effective noise channel, and p_l being the effectiveness factor satisfying

$$\min\left(\lambda\{C_{[\mathcal{E}_l-(1-p_l)\mathcal{I}]}\}\right) = 0,\ 0 \leq p_l \leq 1. \tag{10}$$

We employ the PTM for detailed analytical deductions. PTM is a useful representation for any n-qubit quantum channel A:

$$(\mathcal{R}_{\mathcal{A}})_{ij} = \frac{1}{2^n}\mathrm{Tr}[P_i \mathcal{A}(P_j)], \tag{11}$$

where P_i is the i-th Pauli operator from Pauli group $\mathcal{P}^{\otimes n}$ where $\mathcal{P} = \{I, X, Y, Z\}$. In the PTM representation, the CPTP channel's essential properties of complete positivity (CP) and trace preservation (TP) are reflected as:

- CP constraint: The Choi-Jamiolkowski (CJ) [35, 47] matrix $\mathcal{C}_{\mathcal{A}}$, which can be written in terms of the PTM as

  $$\mathcal{C}_{\mathcal{A}} = \frac{1}{2^{2n}}\sum_{i,j=1}^{d^2}(\mathcal{R}_{\mathcal{A}})_{i,j} P_j^T \otimes P_i, \tag{12}$$

  satisfys $\mathcal{C}_{\mathcal{A}} \geq 0$.

- TP constraint: $(\mathcal{R}_{\mathcal{A}})_{0j} = \delta_{0j}$.

The main purpose of the MND is to isolate the maximal identity component within the noise channel, enabling our model

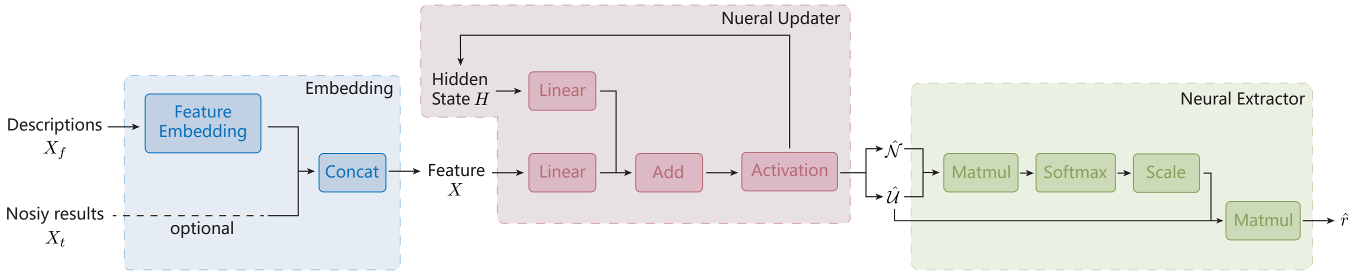

FIG. E1. **The architecture of NNAS**. The architecture mainly consists of three parts. (1) Embedding: We realize the learnable embedding methods to extract relevant information from input and concatenate them together as the feature X. (2) Neural Updater: The recurrent network with hidden state implicitly represent the compressed cricuit and noise matrix, simulates the recursive noise accumulation process. (3) Neural Extractor: The modified attention mechanism effectively surrogate the noisy term.

to more accurately characterize the noise behavior. During decomposition defined in Eq. 9, we constrain the values of p_l to ensure that Λ_l remains a CPTP channel, thus guiding the model to adhere to physical laws. Since the TP constraint $(\mathcal{R}_{(\mathcal{E}_l-(1-p)\mathcal{I})})_{0j} = \delta_{0j}$ holds for any $0 \le p_l \le 1$, the p_l is defined as in Eq. 10 to satisfy the CP constraint.

We illustrate the MND with the typical incoherent Pauli noise channel. For layer l and error rate p, the n-qubit Pauli noise channel is

$$\mathcal{E}_p^l = \sum_{i=0}^{4^n-1} \delta_{i,p}^l \mathcal{P}_i(\cdot), \tag{13}$$

where $\mathcal{P}_i$ is the super-operator of the i-th Pauli operator and $\{\delta_{i,p}^l\}_i$ are the coefficients of Pauli terms which satisfy $\delta_{i,p}^l \ge 0$, $\sum_i \delta_{i,p}^l = 1$. We can easily calculate the effectiveness factor of $\mathcal{E}_p^l$ that $p_l = \sum_{i>0} \delta_{i,p}^l = 1 - \delta_{0,p}^l$, which is the total error probability of the non-trivial Pauli terms. For the incoherent Pauli noise channel, the effectiveness factor p_l is correlated with the fidelity decay rate, which can be efficiently estimated using benchmarking protocols [48–50]. However, in the case of coherent noise channels, where noise is congruent with unitary operations, the effectiveness factor p_l is unity, rendering that the MND doesn't work. Yet our model can still be employed to learn the effectiveness of the noise channel $\mathcal{E}_l$. Furthermore, by employing noise tailoring strategies, such as randomized compiling [51–53], we can transform any CPTP noise channel into an incoherent Pauli noise channel.

Cumulative noise update function. By using MND, the noisy quantum circuit $\tilde{\mathcal{U}}_L$ with L layers,

$$\tilde{\mathcal{U}}_L = \prod_{l=1}^{L} \mathcal{E}^l \circ u_l = \prod_{l=1}^{L} [(1-p_l)\mathcal{I} + p_l\Lambda_l] \circ u_l, \tag{14}$$

can be expressed in a form analogous to the binomial theorem expansion. After merging terms, it reads

$$\tilde{\mathcal{U}}_L = \prod_{l=1}^{L} [(1-p_l)\mathcal{U}_L + [\mathcal{U}_L \mathcal{N}_L \mathcal{U}_L^{\dagger}]\mathcal{U}_L, \tag{15}$$

with the cumulative noise A_L gives,

$$\mathcal{N}_L = \sum_{M=1}^{L} \sum_{k=1}^{\binom{L}{M}} \Big(\prod_{j\in s_M^k} p_j \mathcal{U}_j^{\dagger} \Lambda_j \mathcal{U}_j \Big) \cdot \prod_{j'\notin s_M^k} (1-p_j'), \tag{16}$$

Here s_M^k represents the set of indices for the k-th occurrence of effective noise across M layers.

The iterative update formula of the cumulative noise $\mathcal{N}_l$ in terms of the layer number l is given in Eq. 4, which can be intuitively understood as follows. After introducing the l-th noisy layer $\tilde{u}_l$, the event of the occurrence of effective noise in the circuit $\tilde{\mathcal{U}}_l$ can be delineated into three discrete scenarios:

- The $\tilde{u}_l$ does not introduce effective error, denoted as $(1-p_l)\mathcal{N}_{l-1}$.
- The $\tilde{u}_l$ introduces effective error while the preceding $l-1$ layers remain error-free, characterized as $p_l \prod_{j=1}^{l-1}(1-p_j)a_l$.
- The $\tilde{u}_l$ introduces effective error along with errors occurring in the preceding $l-1$ layers, denoted as $p_l a_l \mathcal{N}_{l-1}$.

B. Detailed construction of physics-inspired neural network

We will next detail the computational procedure of the NNAS model to understand its approach to modeling the noise accumulation process within the multi-layer circuits. The procedure is broadly divided into three key steps:

- Step 1: The uniform neural accumulator recurrently simulate the noise accumulation process.
- Step 2: Recover the layer-wise implicit representation $\mathcal{N}_l$ and $\mathcal{U}_l$ from hidden state H_l in each layer l.
- Step 3: The Attention-assisted extractor combine the information of $\mathcal{N}_l$ and $\mathcal{U}_l$ and calculate r_l with a readout linear layer.

The architecture of NNAS is illustrated in Extended Data Fig.E1 the comprehensive algorithm is outlined in Alg 1.

Algorithm 1: Neural Noise Accumulation Surrogate

Function (G, N) **:**
- G represents the device and circuit specifications, N represents the noisy results, P represents the pre-processed effectiveness factors
- $X_f \leftarrow$ Feature_Embed(G)
- $X \leftarrow$ concat$([X_f, N])$
- **for** $l \leftarrow 1$ *to* L **do**
    - $H_l \leftarrow$ RNN_Cell(X_l) ▷ Step 1
    - Recover $\mathcal{U}_l$, $\mathcal{N}_l$ from H_l ▷ Step 2
    - $A_l \leftarrow$ Softmax$(\frac{\mathcal{U}_l \cdot \mathcal{N}_l^T}{\sqrt{d}})\mathcal{U}_l$ ▷ Step 3
    - $r_l \leftarrow$ Linear_Readout(A_l)
- $y \leftarrow \frac{\mathcal{N}}{(P+r)} + b$ ▷ b is a series of parameter to learn the overall deviation
- **return** y ▷ The prediction results

Feature embedding. The input data typically include specific information of the device and circuit and the observed noisy results. We delve into the embedding layers to extract relevant information from original input data, which aims to enhance the model's ability for recognizing different circuit instances.

We use M variables to describe the specifications of device and circuit, such as quantum system size, type of the circuit, type of the noise, error rate for single gates, error rate for double gates, and circuit parameters. The realization of the feature embedding diverse according to the form of the specification data.

- For single discrete values, formally $G \in \mathbb{N}^{1*1}$, such as quantum number and circuit type, we use a learnable embedding function $E = \mathrm{f(G)} = GW_1$ to embed it, where $W_1 \in \mathbb{R}^{1*1}$ is a learnable scale factor to scale the discrete values into appropriate range. Then we encoded it with linear layer $X^{sd} = \mathrm{g(E)} = EW_2 + b$ where $W_2 \in \mathbb{R}^{1*L}$ and $b \in \mathbb{R}^{1*L}$ are learnable parameters, and we obtain the embedded feature $X^{sd} \in \mathbb{R}^{1*L}$.

- For single continuous values, formally $G \in \mathbb{R}^{1*1}$, such as delta, we use a linear function $\mathrm{E} = \mathrm{f(G)} = GW_1 + b$ to embed it, where $W_1 \in \mathbb{R}^{1*1}$ is a learnable scale factor to scale the discrete values into appropriate range and $b \in \mathbb{R}^{1*1}$ is a learnable offset. Then we encoded it with linear layer $X^{sc} = \mathrm{g(E)} = EW_2 + b$ where $W_2 \in \mathbb{R}^{1*L}$ and $b \in \mathbb{R}^{1*L}$ are learnable parameters, and we obtain the embedded feature $X^{sc} \in \mathbb{R}^{1*L}$.

- For multi-variables with discrete values, formally $G \in \mathbb{N}^{1*s}$, such as observables. Defining s is the size of variables, we transform it into $G^{\mathrm{T}} \in \mathbb{N}^{s*1}$ and fit it with an embedding function $E = \mathrm{f(G)} = GW_1$, where $W_1 \in \mathbb{R}^{s*L}$ is a set of learnable parameters to decompose the impact of G to each circuit layer. Then we use a linear layer to combine the impact of all the variables, namely $X^{md} = \mathrm{g(E)} = E^T W_2 + b$, where $W_2 \in \mathbb{R}^{s*1}$ and $b \in \mathbb{R}^{1*1}$ are learnable parameters. Finally, we transpose it and obtain the embedded feature $X^{md} \in \mathbb{R}^{1*L}$.

Besides, to enhance model robustness, especially with limited descriptive data, we optionally integrate noisy results $X^n \in \mathbb{R}^{1*L}$ with embedded descriptions to extract latent statistical patterns.

All kinds of specifications with different forms are properly embedded with suitable methods to map the global information to each layer. Then they are concatenated together as $X^f \in \mathbb{R}^{M*L}$, where M is the total kind of specific variables and L is the length of circuit.

Uniform neural accumulator simulating the noise accumulation. As $\mathcal{U}_l$ and $\mathcal{N}_l$ are recursively cumulated in equations 14 and 16, we naturally opt for an RNN to simulate their iteration process. Formally, the recurrent process can be expressed as

$$\begin{aligned} \mathcal{N}_l &= \phi_{\mathcal{N}}(X_l, \mathcal{N}_{l-1}, \mathcal{U}_{l-1}), \\ \mathcal{U}_l &= \phi_{\mathcal{U}}(X_l, \mathcal{N}_{l-1}, \mathcal{U}_{l-1}). \end{aligned} \tag{17}$$

With $\mathcal{N}_l$ and $\mathcal{U}_l$ satisfying the cross-spread relation, we utilize a hidden state H_l to implicitly parameterize them. And the recurrent function $\phi_{\mathcal{N}}(\cdot)$ and $\phi_{\mathcal{U}}(\cdot)$ can be implicitly expressed through the recurrent neural network function $\mathcal{F}$, which is formulated as:

$$H_l = \mathcal{F}(X_l, H_{l-1}) = \mathrm{RNN}(X_l, H_{l-1}), \tag{18}$$

The RNN cell consists of two linear projections and an activate operator. Specifically, in each iteration, the hidden state H_l of the circuit layer l is calculated as:

$$H_l = \tanh(\mathrm{Linear}_x(X_l) + \mathrm{Linear}_h(H_{l-1})), \tag{19}$$

where $\tanh$ is an activate operator and the projection $\mathrm{Linear}(\cdot)$ with learnable parameter W and b can be formulated as:

$$\mathrm{Linear}(X) = W \cdot X + b. \tag{20}$$

The hidden state H_l implicitly represents the compressed cross-spread pattern of $\mathcal{N}_l$ and $\mathcal{U}_l$. The hidden state H_l will not be decomposed in the recurrent process to avoid introducing additional accuracy errors and reducing efficiency.

Attention-assisted extractor based on a computationally similar attention core. The Attention mechanism [54] has revealed a promising dependencies alignment ability in sequence learning [55–57], which can be defined as:

$$\mathrm{Attn}(Q, K, V) = \mathrm{Softmax}(\frac{Q \cdot K^T}{\sqrt{d}})V, \tag{21}$$

where d is the scaled factor of data dimension, and Q, K and V are linear projected from input data X.

Though observing the mathematical formulation of noise envelope $R_l = \mathcal{U}_l \mathcal{N}_l \mathcal{U}_l^\dagger$, we introduce the attention mechanism as the neural extractor to surrogate the calculation of R_l. Then, we represent the hidden components of $\hat{\mathcal{N}}_l$ and $\hat{\mathcal{U}}_l$ from H_l in the following way:

$$\begin{aligned} \hat{\mathcal{U}}_l &= W_{\mathcal{U}} \cdot H_l + b_{\mathcal{U}}, \\ \hat{\mathcal{N}}_l &= W_{\mathcal{N}} \cdot H_l + b_{\mathcal{N}}. \end{aligned} \tag{22}$$

Thus, we substitute the new definition into vanilla attention mechanism.

$$A_l = \mathrm{Softmax}(\ (\frac{\hat{\mathcal{U}}_l \hat{\mathcal{N}}_l^{\mathrm{T}}}{\sqrt{d}})\hat{\mathcal{U}}_l\ ), \tag{23}$$

where the hidden states H_l are transformed into $\hat{\mathcal{N}}_l$ and $\hat{\mathcal{U}}_l$ that simulate the $\mathcal{N}_l$ and $\mathcal{U}_l$, respectively. After that, the linear projection is defined as a simple readout layer on A_l.

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DECLARATIONS

Code and data availability

Code and data are available from the corresponding author upon reasonable request.

Author contribution

H.-L. H. initiated the project. H.-L. H., H.-Y. Z. crafted the general workflow for this study. H.-L. H., H.-Y. Z. and W.-S. B. supervised the work. X.-Y. X., X. X. and T.-Y. C. proposed the framework of the Neural Noise Accumulation Surrogate. X.-Y. X. and X. X. conducted the experiments and analyzed the results. X.-Y. X., X. X., H.-L. H. and H.-Y. Z. wrote the paper. All authors contributed to the discussion of the results.

Competing interests

The authors declare no competing interests.

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